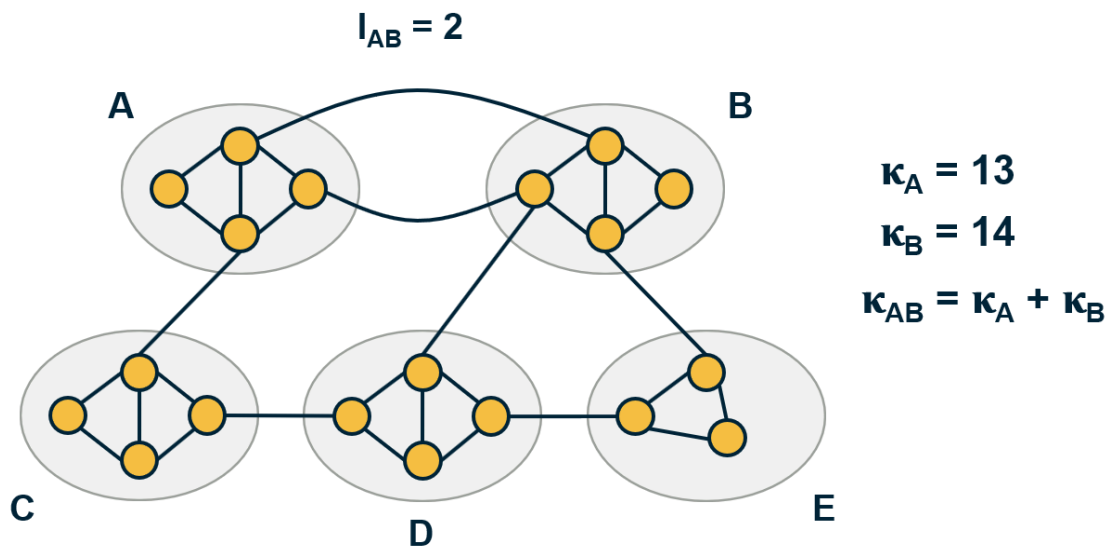


L7: Modularity After Merging Two Communities



In the following pages, we will discuss a couple of algorithms that perform community detection by gradually merging smaller communities into larger communities.

Before we look at these algorithms, however, let us derive a simple formula that shows how the modularity of a network increases when we merge two communities into a new community.

In more detail, suppose that community A has a number of internal links L_A and a total degree κ_A . Similarly for community B .

What happens if we create a new community assignment in which A and B are merged into a single community, call it $\{AB\}$, while all other communities remain the same?

The number of internal links in $\{AB\}$ is $L_{AB} = L_A + L_B + l_{AB}$, where l_{AB} is the number of external links between communities A and B .

The total degree of community $\{AB\}$ is $\kappa_{AB} = \kappa_A + \kappa_B$.

Using the modularity formula we derived earlier, we can now calculate the modularity difference after merging A and B :

$$\Delta M_{AB} = \left(\frac{L_{AB}}{L} - \left(\frac{\kappa_{AB}}{2L} \right)^2 \right) - \left(\frac{L_A}{L} - \left(\frac{\kappa_A}{2L} \right)^2 + \frac{L_B}{L} - \left(\frac{\kappa_B}{2L} \right)^2 \right)$$

All the remaining terms, related to communities other than A and B , have not been changed after the merging and so they cancel out.

After some basic algebra in this expression, we can simplify the modularity difference after merging A and B to:

$$\Delta M_{AB} = \frac{l_{AB}}{L} - \frac{\kappa_A \kappa_B}{2L^2}$$

This is a useful expression, showing that the merging step results in higher modularity only when the number of links between A and B is sufficiently large for the first term to be larger than the second term. And the larger the communities A and B are, the larger the second term is. Otherwise, this merging operation causes a reduction in the modularity of the new community assignment.

Food For Thought

If the previous derivations are not clear, please do the algebra in more detail yourself.